

Food Macro Estimation from Cooking Videos

Foundations of Artificial Intelligence — Final Report

Shravan Chandrachud

1 Introduction

Estimating the nutritional content of a meal from visual information is a challenge at the intersection of computer vision and health informatics. This project builds an end-to-end system that processes cooking videos to identify ingredients and estimate macronutrient content (calories, protein, carbohydrates, fat). The pipeline leverages CLIP (Contrastive Language-Image Pre-Training), a vision-language model that learns transferable visual representations from large-scale image-text pairs. CLIP’s pretrained encoder produces rich embeddings that rival task-specific supervised models on food recognition benchmarks. However, CLIP’s standard softmax-based inference assumes one class per image, which fails for cooking scenes containing multiple ingredients simultaneously. We address this by replacing softmax with per-class sigmoid activation and training a lightweight MLP classification head on frozen CLIP ViT-L/14 embeddings, enabling multi-label detection. Detected ingredients are confirmed by the user (who provides quantities), then mapped to USDA nutritional data and summarized by a locally-hosted LLM into a structured macro breakdown.

2 Methods

2.1 Dataset

The dataset comprises approximately 16,990 images across 6 parent classes and 20 subclasses sourced from Fruits-360, VegFru/Veg200, a Rice Image Dataset, a Seafood dataset, and egg/oil collections. Data was split 70/15/15 into train (11,885), validation (2,541), and test (2,564) sets with a fixed random seed.

Table 1: Dataset distribution across splits.

Class	Subclasses	Train	Val	Test	Total
Egg	1	113	24	25	162
Fruits	5	5,994	1,282	1,291	8,567
Oil	1	685	146	148	979
Rice	5	1,750	375	375	2,500
Seafood	3	892	190	194	1,276
Vegetables	5	2,451	524	531	3,506
Total	20	11,885	2,541	2,564	16,990

2.2 Multi-Label Classifier

Frozen CLIP ViT-L/14 (OpenCLIP, LAION-2B pretrained, 768D embeddings) is used as a feature extractor. A two-layer MLP head ($768 \rightarrow 256 \rightarrow 20$) with ReLU, dropout (0.3), and per-class sigmoid activation is trained on precomputed embeddings using BCEWithLogitsLoss with per-class positive weighting ($w_j = N_{\text{neg},j}/N_{\text{pos},j}$) to handle class imbalance. Per-class decision thresholds are tuned on the validation set by sweeping 0.75–0.95, selecting the F1-maximizing threshold per class. The 0.75 minimum floor addresses the domain gap between clean training images and complex cooking video frames where low thresholds caused pervasive false positives.

2.3 Video Pipeline

Keyframes are extracted using PySceneDetect’s AdaptiveDetector for scene-cut boundaries combined with uniform 0.5 FPS sampling and Laplacian blur filtering. Each keyframe is embedded through the frozen CLIP encoder and classified by the MLP head, with multiple ingredients detectable per frame. The user confirms detected ingredients and provides quantities. Confirmed ingredients are matched against an offline USDA FoodData Central SQLite database (~7,800 foods) via fuzzy string matching (RapidFuzz), then sent with USDA reference macros to a locally-hosted Qwen 2.5 3B LLM (via Ollama) which generates a structured JSON nutritional summary enforced by a Pydantic schema. CLIP and the LLM run sequentially on GPU to stay within 6GB VRAM.

3 Results

3.1 Classifier Performance

The MLP achieves 99.2% macro F1 and 99.6% micro F1 on the held-out test set across all 20 subclasses. All classes exceed 0.96 F1, with 8 classes achieving perfect 1.000.

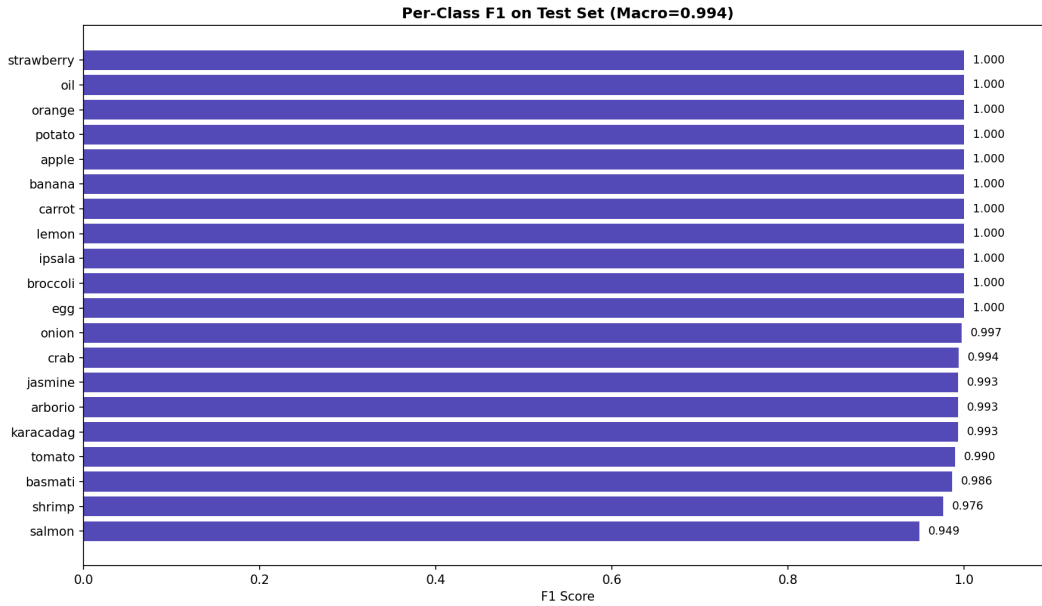


Figure 1: Per-class F1 scores on the test set.

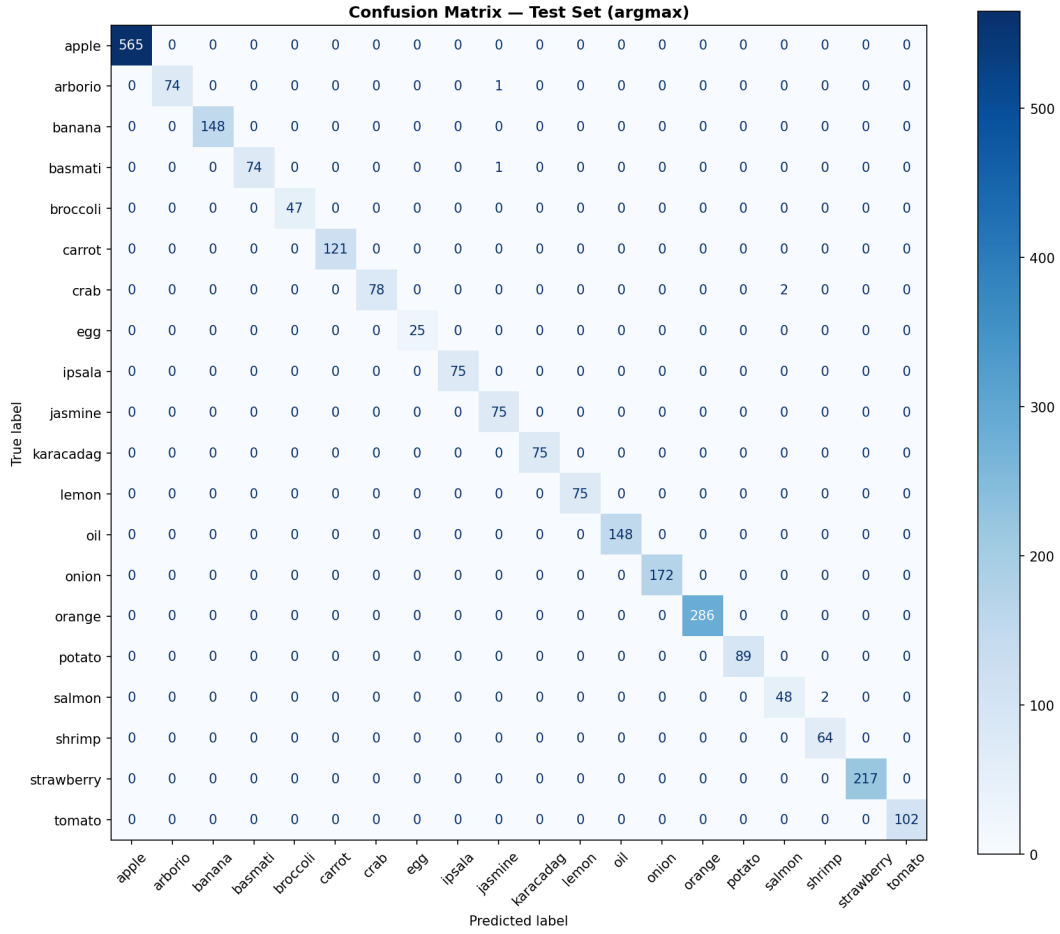


Figure 2: Confusion matrix on the test set. Nearly perfectly diagonal with minor confusions between rice varieties and seafood subclasses.

3.2 Embedding Visualizations

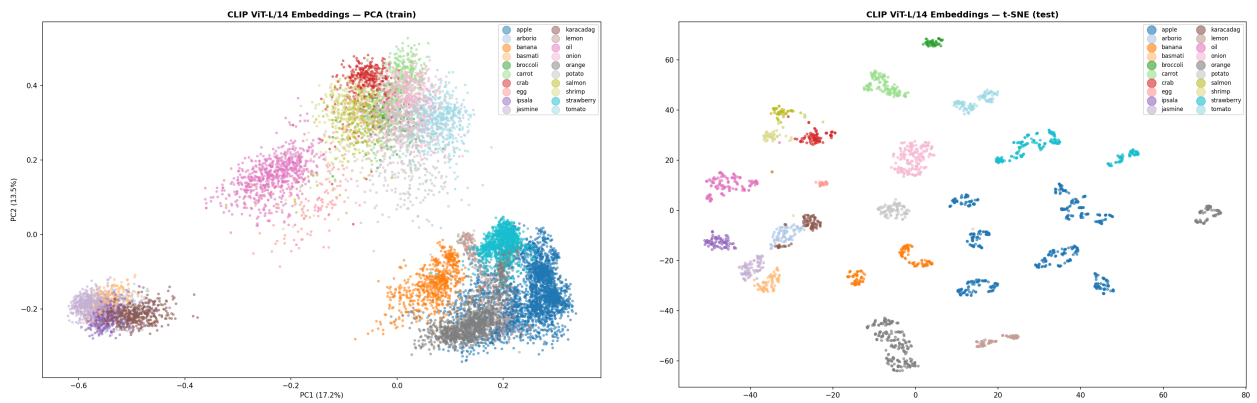


Figure 3: CLIP ViT-L/14 embedding space. All 20 classes form well-separated clusters.

3.3 Threshold Calibration and Ablation

Thresholds optimized without a floor produced severe false positives on video frames (oil and egg detected in every frame). Raising the floor to 0.75 eliminated these while maintaining 99.2% test macro F1, demonstrating the domain gap between training data and real cooking scenes.

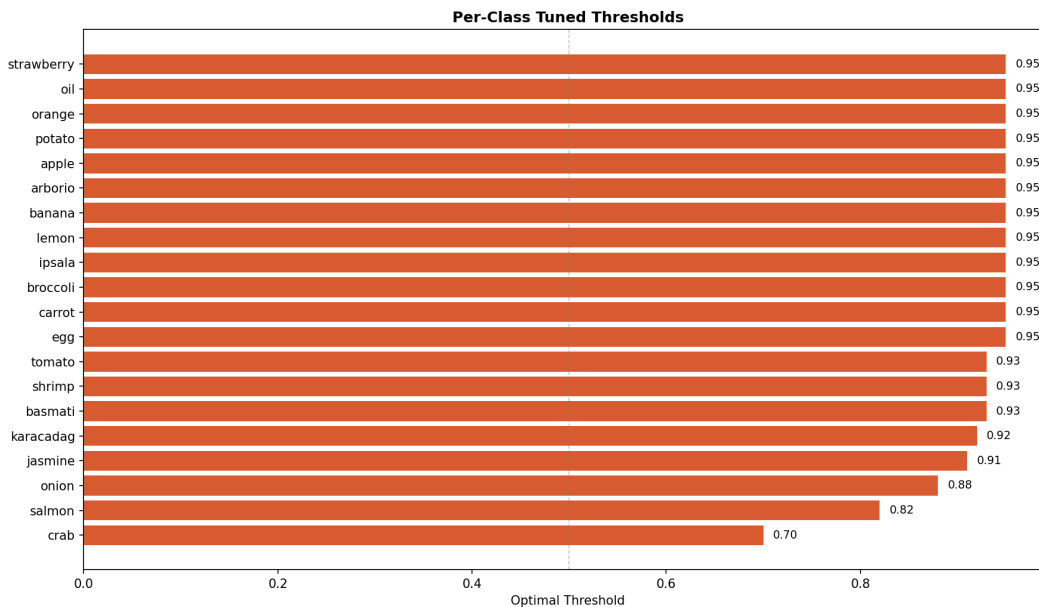


Figure 4: Per-class tuned thresholds with 0.75 minimum floor.

3.4 Video Pipeline Demo

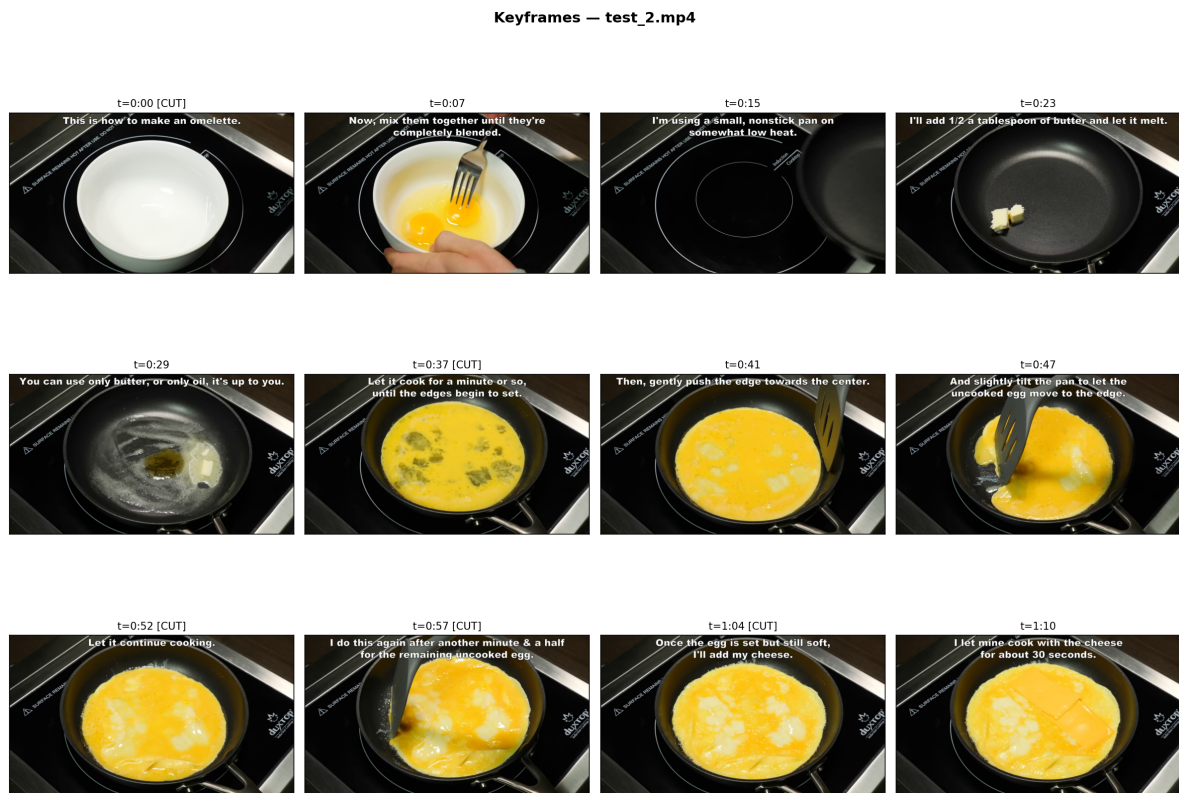


Figure 5: Sample keyframes from test video 1. [CUT] marks scene-change boundaries.

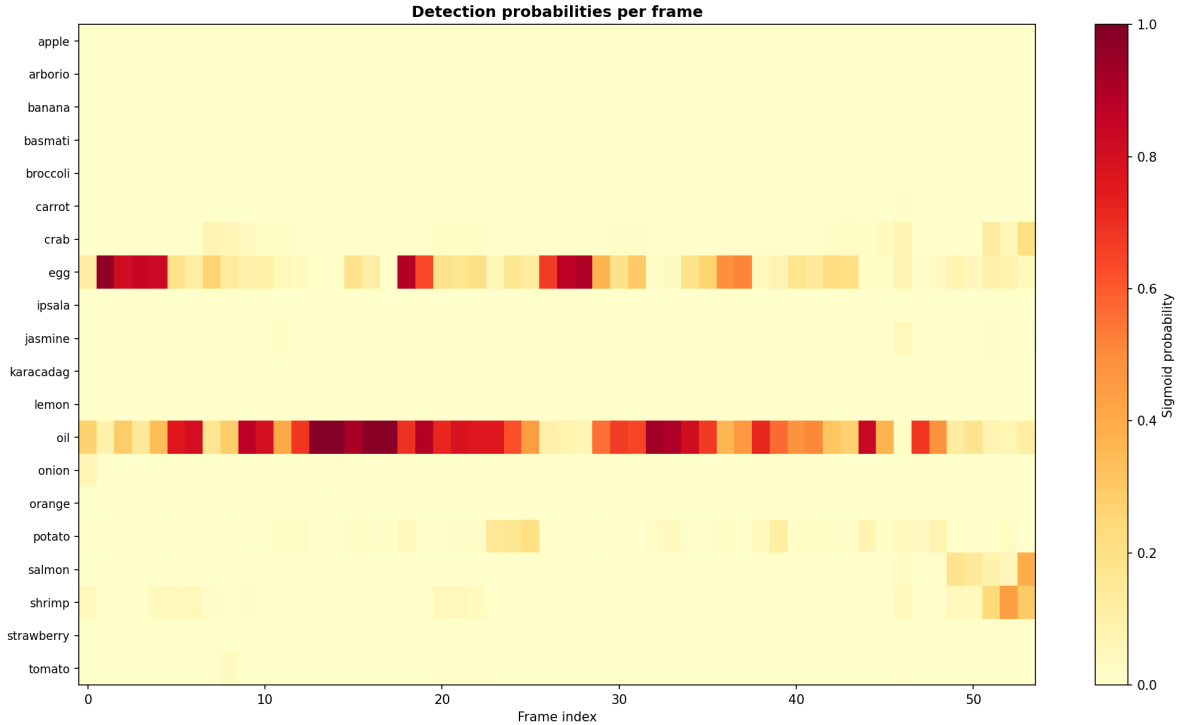


Figure 6: Detection probability heatmap. Rows are ingredient classes, columns are keyframes.

4 Conclusions

The frozen CLIP ViT-L/14 encoder paired with a lightweight MLP head achieves 99.2% macro F1 across 20 ingredient classes, confirming that pretrained vision-language models provide excellent food recognition features without encoder fine-tuning. The sigmoid + BCEWithLogitsLoss formulation correctly handles multi-ingredient scenes, and per-class threshold calibration with a 0.75 floor mitigates the domain gap between clean training images and complex video frames. The human-in-the-loop quantity confirmation is a pragmatic design choice that avoids the unsolved problem of visual portion estimation.

The primary weakness is the domain gap itself — the model was trained on isolated ingredient images but deployed on cluttered cooking scenes. While the threshold floor addresses this heuristically, a more robust solution would involve fine-tuning on annotated cooking video frames or integrating object detection (e.g., YOLO-World) for region-level classification. Compared to end-to-end systems like Nutrition5k or FoodLMM, our approach is significantly simpler and more resource-efficient, running entirely on a single consumer GPU, though it requires user input for quantity estimation. Compared to pure zero-shot CLIP classification, the trained MLP with threshold calibration provides substantially more reliable detections in practice.

5 Works Cited

References

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